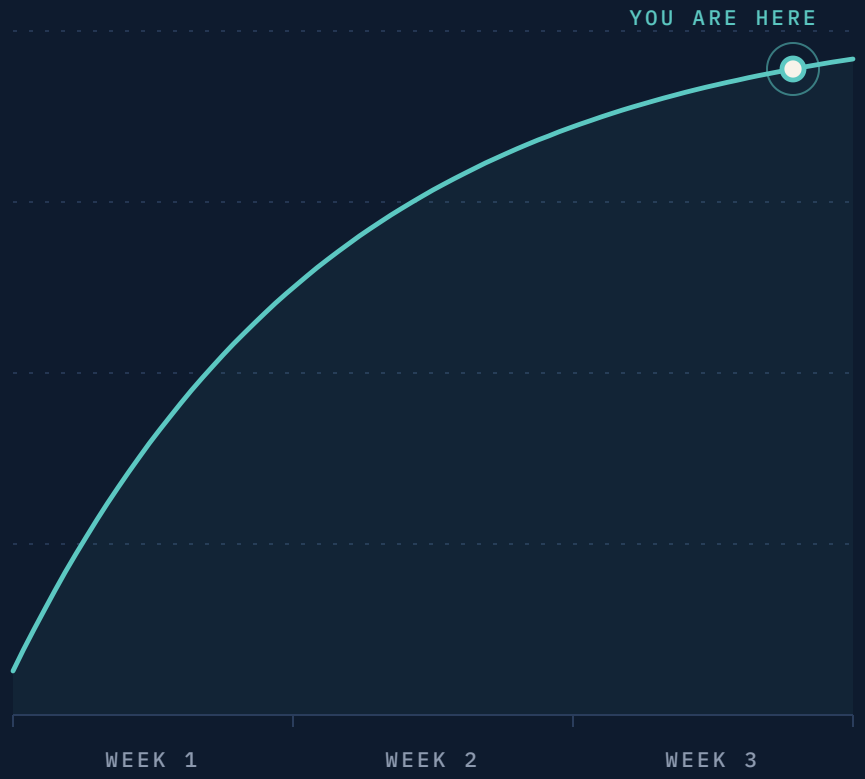
WEEK 3 · DAY 4 · OVERVIEW

SHAP Values



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ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

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Problem Statement

BUSINESS CONTEXT

Days 11 to 13 settled a model for the colleague's Polish file: LightGBM at its defaults, unweighted, at a fold average precision of 0.7783.

The file holds financial statements of Polish companies, the bankrupt ones analysed from 2000 to 2012 and the operating ones from 2007 to 2013 (dataset page), so it describes one country in one period.

A credit committee now asks the desk two things: for any one company, how does the model arrive at its number, and which columns does it lean on overall?

SHAP values answer both, by splitting each company's output into one contribution per column.

A SHAP value is a column's contribution to this company's log-odds, relative to the model's average log-odds, never a reason, a cause, or the effect of changing a ratio.

A model's output here is the model's probability of bankruptcy in this file, never a credit view on any company.

OBJECTIVES

Compute the SHAP values of a fitted booster, check that they add up to its log-odds, and turn log-odds into a probability.

Read a waterfall, a decision-path chart, a beeswarm and a dependence plot, and compare SHAP values with gain and split counts.

See what a near-copy column does to SHAP values, and describe companies with models that never saw them.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 4

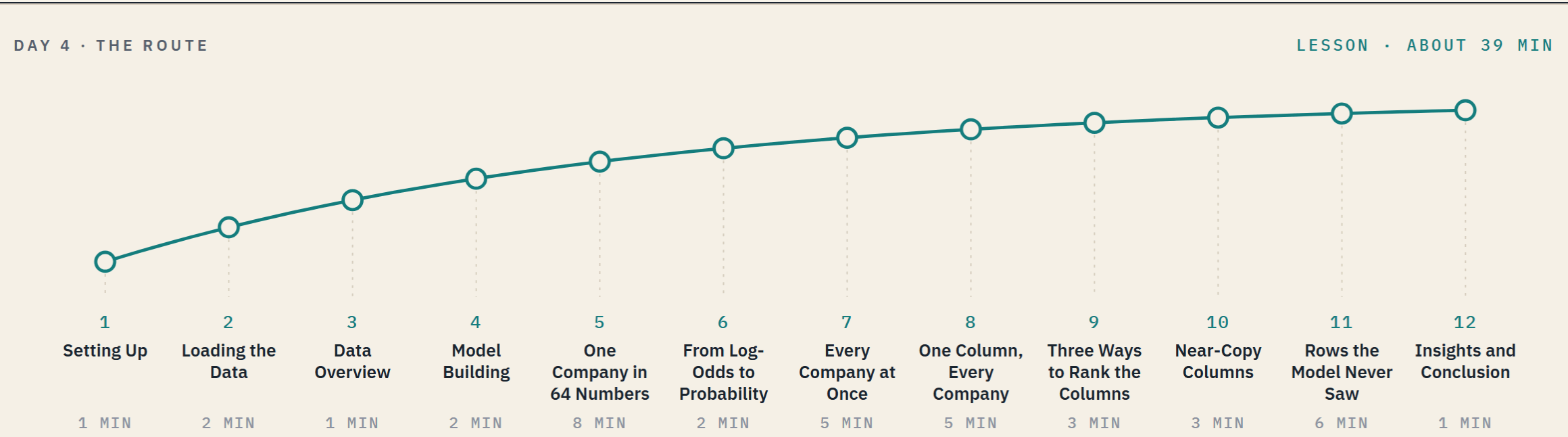


FIGURE 1 Twelve sections, in order, with about how long each takes.

Time. About 39 minutes for the lesson, then about 12 for the four exercises.

Setting Up

THE LIBRARIES

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from ipywidgets import interact
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_validate
from sklearn.inspection import permutation_importance
from lightgbm import LGBMClassifier
from xgboost import XGBClassifier
import shap
```

THE SEED

```
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
```

THE COLUMNS MET TODAY

- **`Attr1`** to **`Attr64`**, the page's X1 to X64 in order, are the 64 features. Those met by name today:
- **`class`**: 1 if the company went bankrupt within a year, **the target**

Rebuild Course 4's training rows and five folds.

Loading the Data

READ THE FILE

```
bankrupt = pd.read_csv(DATA + "polish_bankruptcy_5year.csv")
TARGET = "class"
FEATURES = [column for column in bankrupt.columns if column != TARGET]
print(len(bankrupt), "statements,", len(FEATURES), "features")
```

```
5910 statements, 64 features
```

REBUILD COURSE 4'S ROWS

```
unique = bankrupt.drop_duplicates()
X, y = unique[FEATURES], unique[TARGET]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=SEED)
print("Training:", len(X_train), "rows,", y_train.sum(), "bankrupt.  Test:", len(X_test), "rows,", y_test.sum(), "bankrupt")
```

```
Training: 4680 rows, 326 bankrupt.  Test: 1170 rows, 82 bankrupt
```

Data Overview

COUNT THE SHARE AND THE GAPS

```
print("Bankrupt:", round(y_train.mean() * 100, 2), "percent")
print("Columns with a gap:", X_train.isna().any().sum(), " Companies with at least one:", X_train.isna().any(axis=1).sum())
```

```
Bankrupt: 6.97 percent
Columns with a gap: 48  Companies with at least one: 2248
```

FIVE FOLDS

```
folds = StratifiedKFold(n_splits=5, shuffle=True, random_state=SEED)
```

The rows go in as they are.

Model Building

SCORE BY FOLDS

```
model = LGBMClassifier(random_state=SEED, n_jobs=1, verbose=-1)
scores = cross_validate(model, X_train, y_train, cv=folds, scoring="average_precision", n_jobs=1)
print("Folds:", scores["test_score"].round(4), " Mean:", round(scores["test_score"].mean(), 4))
```

```
Folds: [0.7077 0.7988 0.7434 0.8013 0.8405]  Mean: 0.7783
```

FIT ONCE

```
model.fit(X_train, y_train)
proba = model.predict_proba(X_train)[:, 1]
print("Average output:", round(proba.mean(), 4), " Bankrupt share:", round(y_train.mean(), 4))
```

```
Average output: 0.0697  Bankrupt share: 0.0697
```

One fitted model to describe; from here no company is scored, only described.

One Company in 64 Numbers

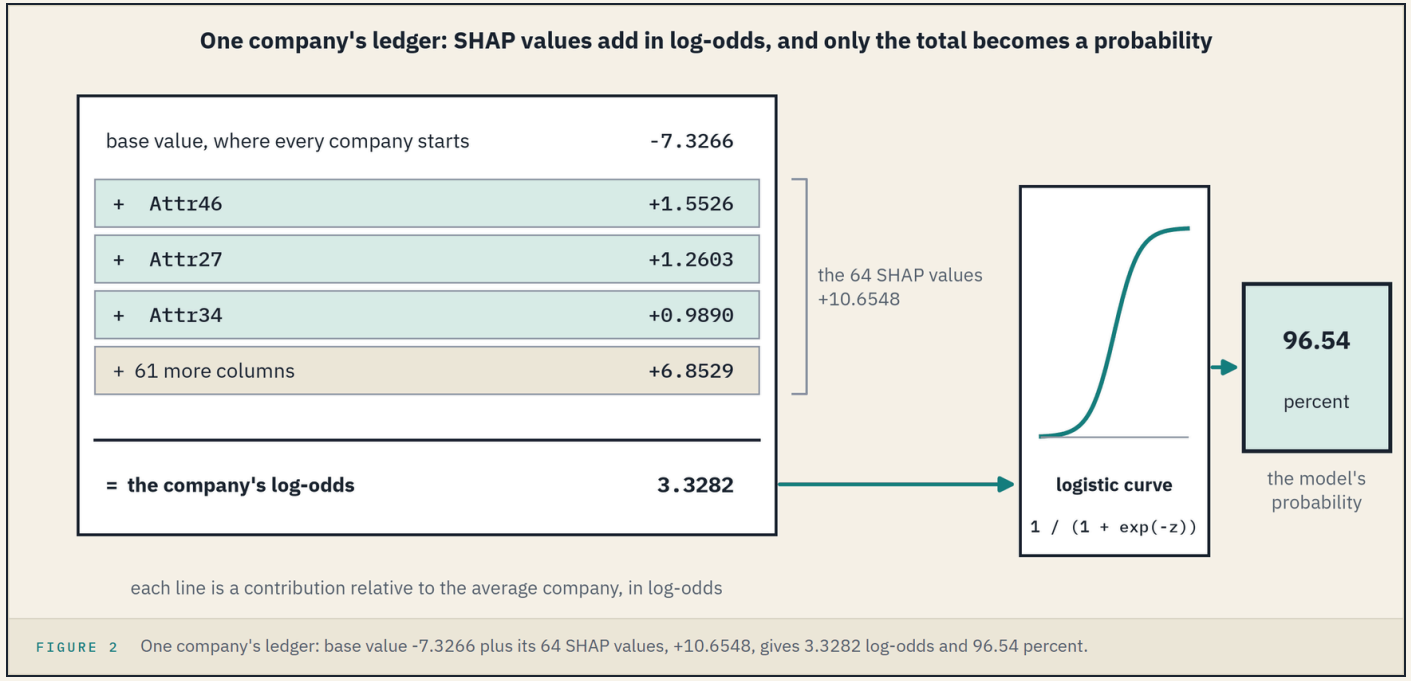


FIGURE 2 One company's ledger: base value -7.3266 plus its 64 SHAP values, +10.6548, gives 3.3282 log-odds and 96.54 percent.

CHECK EVERY ROW

```
rebuilt = base + shap_values.values.sum(axis=1)

Largest gap over 4,680 companies: 1.6e-13 log-odds
```

ITS LARGEST CONTRIBUTIONS

	VALUE	SHAP VALUE
Attr46	0.0420	1.5526
Attr27	-1.7329	1.2603
Attr34	0.5655	0.9890
Attr39	-0.0902	0.9603
Attr38	-1.0383	0.9485
Attr35	-0.0926	0.7060

The base value plus a company's SHAP values gives its log-odds to the 13th decimal place, for every company: additivity.

From Log-Odds to Probability

THE QUESTION

Attr46 adds 1.5526 log-odds to this company.

Q. In percentage points of probability, what is that step worth counted first, starting from the base value, and counted last, ending at the company's log-odds? Write down two numbers.

COUNT THE STEP TWICE

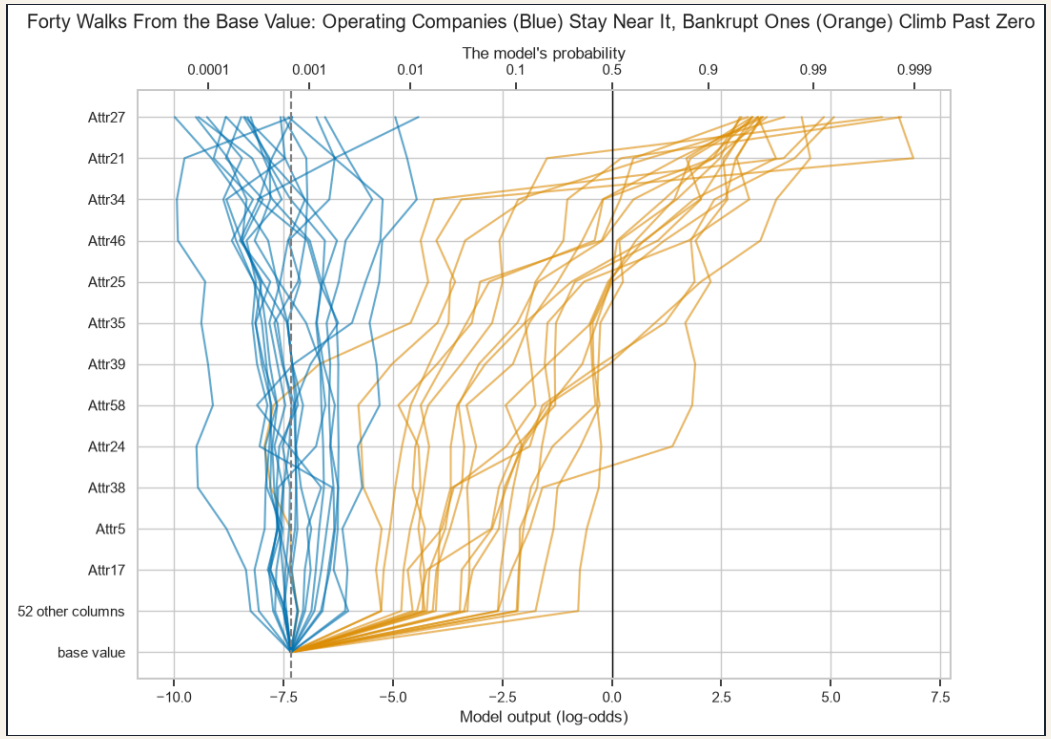
```
step = row.loc["Attr46", "SHAP value"]
first = (logistic(base + step) - logistic(base)) * 100
last = (logistic(log_odds[company]) - logistic(log_odds[company] - step)) * 100
```

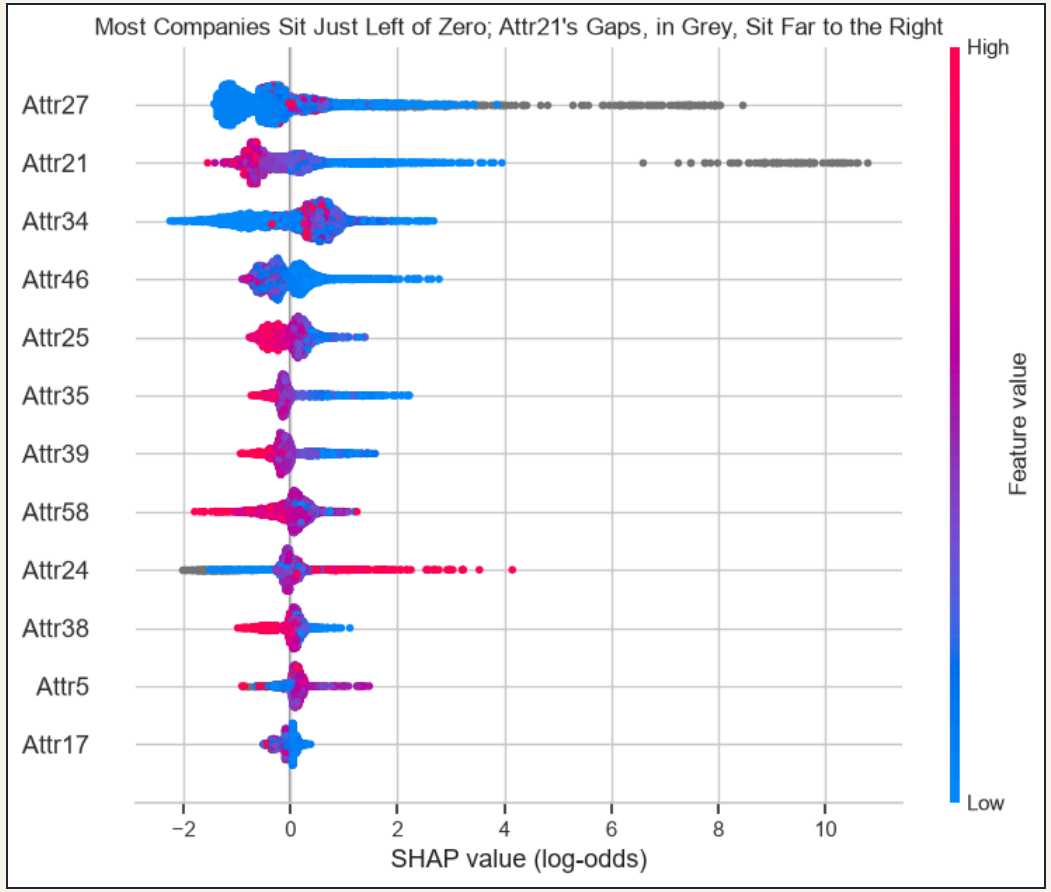
THE LOGISTIC CURVE

```
def logistic(z):
    return 1 / (1 + np.exp(-z))
```

Contributions are stated in log-odds; only a company's total is turned into a probability.

Every Company at Once





The model leans on different columns for different companies; one column at a time shows how.

One Column, Every Company

THE QUESTION

Attr27 is operating profit over financial expenses: below 0 the company loses money before interest, above 1 its operating profit covers its financial expenses.

Q. Sketch **Attr27**'s SHAP value (up) against **Attr27** (across), from deeply negative to very large.

YOUR TURN: ANY OF THE TOP TEN

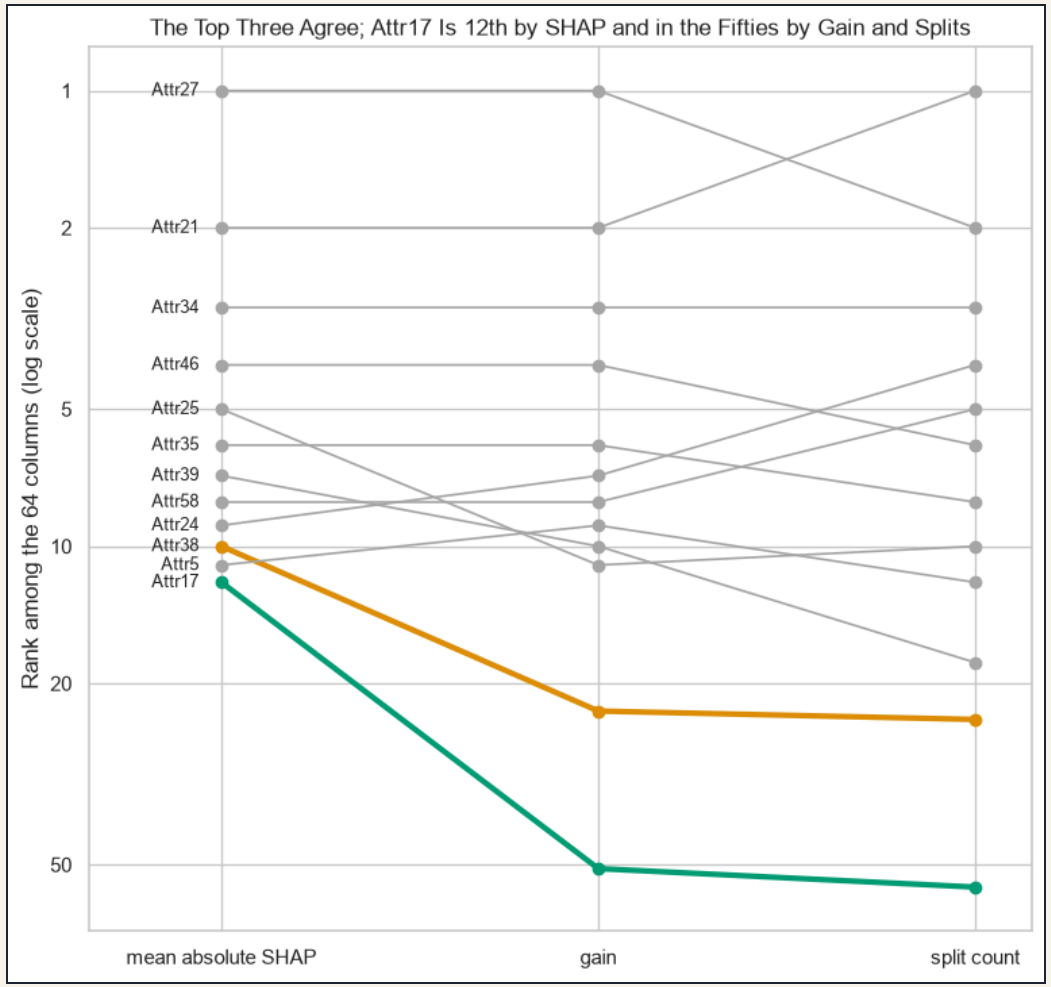
```
interact(draw_dependence, column=list(mean_abs.index[:10]));
```

A COLUMN'S GAPS

```
Gaps: 79  Bankrupt among them: 77  Average SHAP value: 9.2526  Lowest: 6.5679
With a value: average SHAP value -0.1773
```

A gap can carry the largest contribution in the model; a dependence plot shows the model's use of a column, an association in this file.

Three Ways to Rank the Columns



RANKS OF THE TOP COLUMNS

	ATTR27	ATTR21	ATTR34	ATTR46	ATTR25	ATTR35
SHAP	1	2	3	4	5	6
gain	1	2	3	4	11	6
splits	2	1	3	6	10	8

WHERE ATTR17'S SPLITS SIT

```
Splits: 13  At depth 2: 10  Companies reaching those: 2316 or more
```

Each measure answers its own question: how much a column moves the outputs, how much it improved the fit, how often it was used. Quote the one you mean.

Drop a near-copy column and refit.

Near-Copy Columns

THE QUESTION

Q. Refit the model without **Attr17**. What mean absolute SHAP value will **Attr2**, its near-copy, get? Write down a number.

ARE THEY COPIES

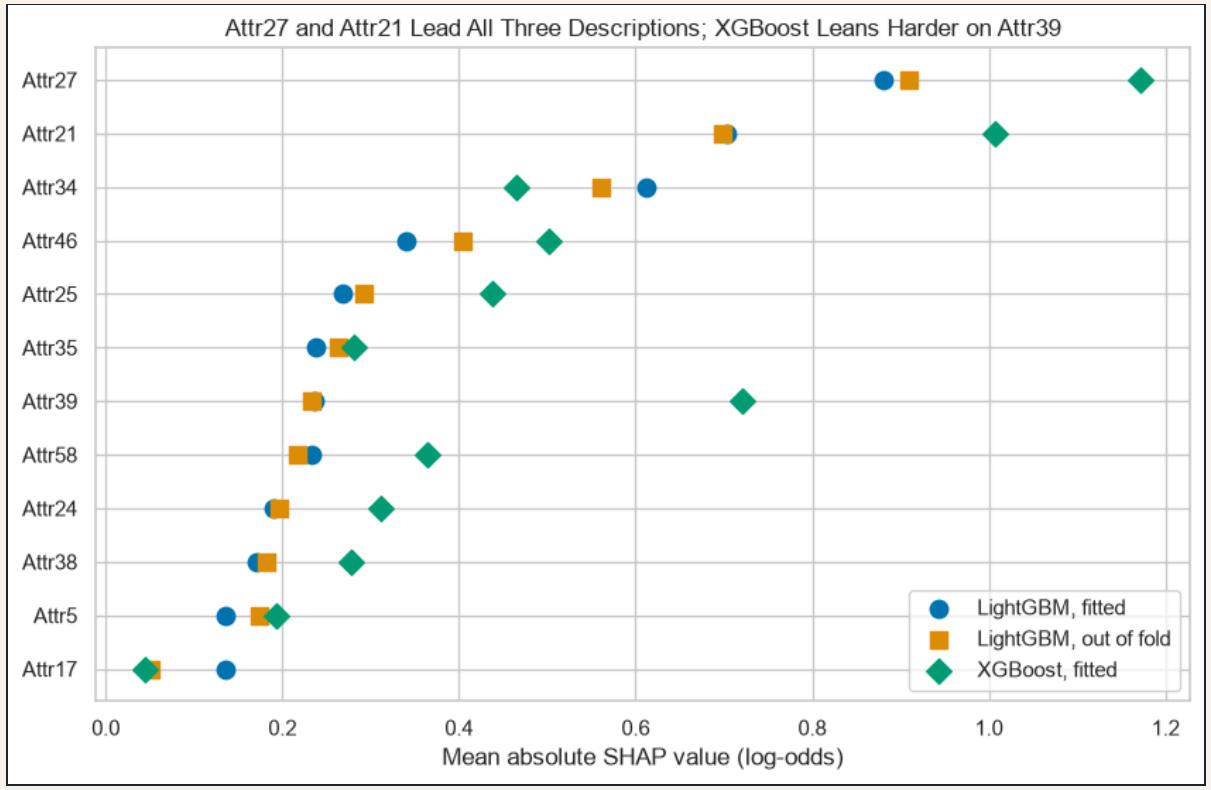
```
both = X_train[["Attr2", "Attr17"]].dropna()
```

```
Companies with both: 4666  Reciprocal within 1 percent: 4665
```

REFIT WITHOUT ATTR17

```
X_no17 = X_train.drop(columns="Attr17")
model_no17 = LGBMClassifier(random_state=SEED, n_jobs=1, verbose=-1).fit(X_no17, y_train)
```

Rows the Model Never Saw



THE QUESTION

Q. What log-odds does the fold model that never saw it give? Write down a number.

DO THE SUMS STILL HOLD

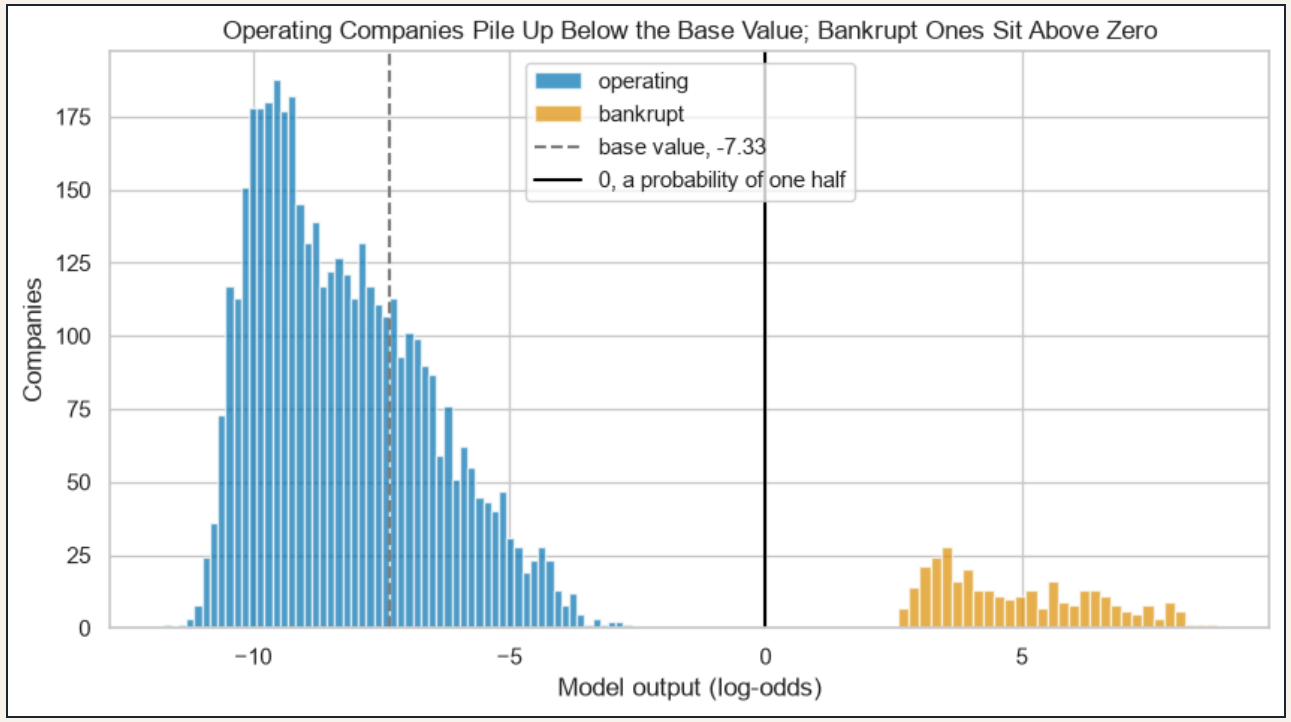
```
Largest gap: 2.2e-12 log-odds
```

A SECOND BOOSTER

```
Stored as: float32  Largest gap: 7.6e-06 log-odds
Top ten shared with LightGBM: 9
```

The same ten columns in the same order: company by company the stories move, but what the model leans on across the file holds.

Insights and Conclusion



WHERE THE COMPANIES SIT

```
Below the base value: 3105  Median log-odds: -8.3174
Average log-odds: operating -8.242  bankrupt 4.902
```

The capstone describes its chosen model with SHAP values on the Taiwan file.

You Can Now

- Compute SHAP values with **`shap.TreeExplainer`** and check that they add up to a booster's log-odds.
- Turn log-odds into a probability, and say why contributions stay in log-odds.
- Read a waterfall, a decision-path chart, a beeswarm, cohort bars and a dependence plot.
- Compare mean absolute SHAP values with gain and split counts, and spot a near-copy column.
- Describe companies with fold models that never saw them.

NEXT

WEEK 3 · DAY 5

Course 5 Capstone

The capstone describes its chosen model with SHAP values on the Taiwan file.